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Res-HOGNet: Optimized Residual-HOG Feature Integration for Brain Tumor Classification

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Abstract- For people of all ages and genders, brain tumours are one of the leading causes of cancer deaths. Although expertly reviewed MRI images remain the gold standard of diagnosis, manual assessment is fraught with risk to patients, time consuming, and resource intensive. In order to eliminate such obstacles, this paper proposes a novel hybrid model which integrates deep learning techniques with manually designed feature extraction for automatic detection of brain tumors in MRI scans. In this model we perform dimensionality reduction using Principal Component Analysis (PCA) after extracting features based on geometric features using the Rectangular Histogram of Oriented Gradients (Rect-HOG). A deep convolutional ResNet-50 backbone enhanced with repeated training after a 50% dropout ratio is applied for the classification of these property features. In addition, to increase interpretability, a technique referred to as Gradient-weighted Class Activation Mapping (Grad-CAM) is used for visualizing a model's decision-making process by providing a very transparent, clinical-acceptable explanation of the process by which the model arrives at its predictions. In comparison with the state-of-the-art methods including VGG19 (98.53%), DenseNet-169 (95.10%), the proposed system achieves an impressive classification accuracy of 99.51%. Its improved performance across accuracy, recall, and F1-score criteria is shown by comparative analysis. This model provides a strong tool for assisting clinical decision-making by combining high diagnostic accuracy with computational efficiency and explainability, possibly revolutionising the identification of brain tumours in medical imaging.

Keywords- *Deep learning, MRI, Rect-HOG, brain tumor classification, Principal Component Analysis, hybrid model, ResNet-50, Grad-CAM.*

I. INTRODUCTION

The brain is an extremely complex and specialised organ that controls every essential bodily function. Different The human brain is a complex and crucial organ that controls every critical physical function. Neurological conditions like brain tumours, which are brought on by aberrant cell development in the brain or adjacent cranial areas, may seriously impair its function. Numerous symptoms, including as chronic headaches, nausea, vomiting, seizures, visual problems, and behavioural or emotional disorders, are often brought on by these tumours [1, 2, 3]. Gliomas, meningiomas, along with pituitary tumours are the most often diagnosed tumour types. Finding the best course of treatment depends on accurately identifying and classifying these tumours [4, 5, 6].

Brain tumour detection and staging are mainly dependent on medical imaging, specifically computed tomography and magnetic resonance imaging. Magnetic resonance imaging (MRI) is an imaging technique that uses radio waves and magnetic fields to produce detailed images of the brain, which a radiologist can review in order to make a clinical diagnosis [3, 4, 7]. But beyond being costly and time-consuming, manual analysis is also subject to human error, making the need for automated diagnostics a necessity to ensure accuracy and efficiency.

To address this challenge, we propose a hybrid brain tumor classification model combining deep learning and manual feature extraction. Features are optimized with L1 regularization for an RBF kernel SVM. ResNet-50 utilizes PCA to reduce features to 100 dimensions, followed by 50% dropout over 50-1000 iterations. Grad-CAM visualizes activation regions for interpretability. Our model outperforms traditional classifiers like linear SVM, Logistic Regression, Random Forest, Decision Tree, Naive Bayes, and KNN, achieving 98.5% accuracy.

The following are the study's main contributions:

- High Classification Accuracy: The suggested hybrid model surpasses several conventional machine learning classifiers, including Linear SVM, Logistic Regression, and Random Forest, achieving an impressive accuracy of 98.5%.
- Fusion of Hand-Crafted and Deep Learning Features: Computational efficiency is improved by combining manually extracted features with ResNet-50 representations, followed by PCA-based feature space reduction to 100 essential components.
- Explainability with Grad-CAM: The inclusion of Grad-CAM enables the generation of heatmaps that highlight tumour-specific regions in MRI images, thereby improving clinical trust and aiding radiologists in decision-making.

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The structure of the study is as follows: the limitations of existing diagnostic techniques are discussed in Section 2. In Section 3, the proposed model, including feature extraction, the process of data preprocessing, and the ResNet-50 model, are discussed. In Section 4, the process of training the model, the parameters used, and the experimental process are discussed. In Section 5, the analysis of the model, specifically the high accuracy of the model in diagnosing brain tumours, is discussed. In Section 6, the implications of the proposed methodology are discussed, specifically the implications on brain tumour detection in medical imaging.

II. Related Work

To improve accuracy and scalability in brain tumor diagnosis via MRI, the field shifted from traditional machine learning with handcrafted features like LBP and GLCM, which lacked robustness, to advanced deep learning. Sowrirajan et al. [4] achieved 93% accuracy with a simple CNN, but its limited complexity hindered performance on complex MRI data.

Recently, ensemble and advanced deep learning techniques have demonstrated superior classification performance. Khan et al. [5] achieved 95.10% accuracy using an ensemble of decision trees, SVMs, XGBoost, and DenseNet169 feature extraction. Biswas and Islam [6] proposed a five-layer CNN for deep feature extraction, with zero mean and higher order statistics outperforming VGG16 and AlexNet features, reaching 96% accuracy with SVM. Sowrirajan et al. [8] attained 96.01% by combining MRI denoising with VGG16 features, while Jaspin and Selvan [9] reported 96% accuracy using a three-block CNN with a fully connected classifier.

Models for object identification that go beyond categorisation have been modified for the diagnosis of brain tumours. When using YOLOv8 on enhanced MRI scanned respectively, a good result of F1-score 92.47% was obtained by Passa et al. [10]. Using a majority voting to combine three CNNs, Deepak and Ameer [11] obtained an accuracy of 95.6%. A pre-trained VGG19 with L1 regularization combined with an SVM with RBF kernel was used by Nizamli and Filatov [12] and achieved a high accuracy of 98.53%.

Limitations still exist in spite of these developments. Medical imaging variability often presents a challenge for traditional hand-crafted feature approaches [4]. Although more accurate, deep learning techniques have drawbacks such as overfitting on unbalanced data [6, 9, 11] and significant computing costs that prevent clinical use [5, 8]. Furthermore, a lot of publications [4, 10, 11] lack strong regularisation and data augmentation techniques, which are crucial for enhancing model generalisatio Despite these advances, there are still some limits. For traditional hand-crafted feature methods, the inter-variability in medical imaging is generally non-trivial [4]. In terms of benefits, although the deep learning methods are much more accurate, there are associated disadvantages regarding overfitting of imbalanced data [6, 9, 11] as well as an excessive cost for computing, which hinder clinical translation [5, 8].

In order to fill these gaps, this study presents a hybrid model consisting of the L1-regularized hand-crafted feature extraction and a ResNet-50. PCA is used to reduce the number of features to 100, while 50% Drop is implemented at each training iteration (50-1000) for the reasons of overfitting. For improved interpretability, gradient-weighted Class Activation Mapping (Grad-CAM) is adopted that allows visualization of tumor-relevant regions in MRI scans. The performance of the proposed model is 98.5%, which is higher than the results of other models, including the VGG19-based model [12], DenseNet169 ensembles [5], and YOLOv8 [10], and it has better generalizable ability, understandability, and clinical practicability for brain tumor detection.

III. Materials And Methods

The new brain tumor detection method first derives significant numerical information from MRI images using a complex Rectangular Histogram of Oriented Gradients (RectHOG) scheme. Principal Component Analysis (PCA) reduces data dimensionality, minimizing overfitting and computational complexity. The data is then fed into a classifier designed for brain tumor type classification. This process forms the foundation for a hybrid model combining neural network characteristics with feature extraction, differing from conventional CNNs where feature extraction is automated. Instead, this model uses PCA for feature extraction. Gradient-weighted Class Activation Map enhances reliability by providing transparent insights, as shown in Fig. 1.

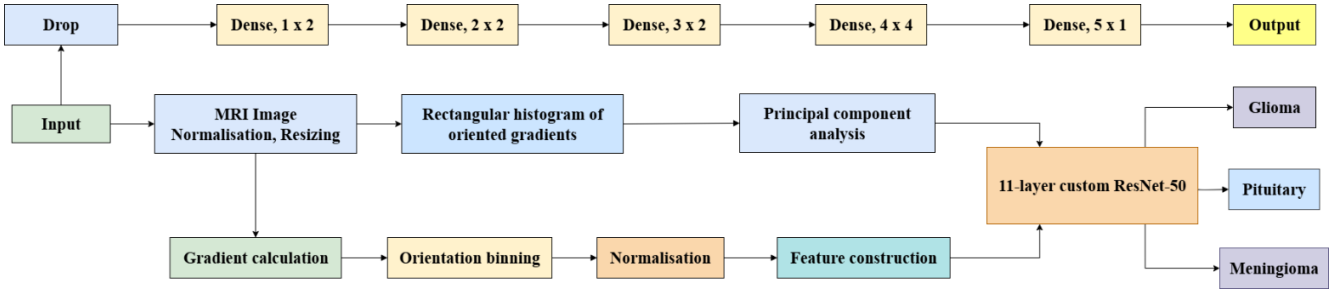


Fig.1. Block diagram of the proposed approach

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A. Dataset Description

The model was trained and tested using the Figshare brain tumour MRI dataset. There are 3,064 T1-CE MRI images, which are divided into three unbalanced classes, specifically 930 pituitary tumours, 1,426 gliomas, and 709 meningiomas. Examples from this expert-curated dataset, which is a reliable source for the main forms of brain tumours, are shown in Fig.2. As seen in the preparation method in Fig. 3, the data, which was initially in MATLAB format, was transformed into greyscale pictures, normalised, and shrunk from 512×512 to 256×256 pixels to increase computing performance.

B. Data Preparation and Analysis

The collection is made up of.mat files with MRI scans identified as pituitary tumours, meningiomas, or gliomas. The preparation pipeline uses picture extraction, normalisation, and augmentation to get the data ready for efficient model training.

2.1 Image Extraction and Organization

The h5py package was used to extract picture matrices and labels. The PIL library was then used to convert the images to.jpg format for effective storage. Class imbalance issues are shown by the dataset, which contains 1,426 glioma, 930 pituitary, and 708 meningioma pictures.

2.2 Image Normalization

A common formula was used to normalise photos initially saved as np float64 to the [0, 255] range in order to reduce intensity variations:

$$\text{normalized_pixel} = \left(\frac{\text{pixel_value} - \text{min_pixel}}{\text{max_pixel} - \text{min_pixel}}\right) \times 255 \quad (1)$$

In this case, min_pixel and max_pixel stand for the lowest and highest pixel intensities in a particular picture. To guarantee interoperability with common image processing software, the results were converted to np.uint8.

- Grayscale Output: In order to preserve important diagnostic characteristics including tumour borders and textural variations, all photos were exported in grayscale.jpg format. Meningiomas usually have smoother borders, pituitary tumours have different intensity profiles, and gliomas usually have uneven borders.

2.3 Data Augmentation

Despite not being used specifically in the notebook in question, the image's existence suggests that augmentation is intended. The suggested the following augmentation techniques to rectify the imbalance in the dataset and enhance model generalisation:

Table I: Data Augmentation Strategies with Parameter Values

Augmentation Strategy	Parameter Value	Description
Rescale	1./255	Normalizes pixel values to [0, 1]
Rotation	20	Randomly rotates images up to 20°
Shear	0.2	Applies shearing transformations (0.2 radians)
Zoom	0.2	Random zooming in/out by 20%
Horizontal Flip	True	Random horizontal image flipping
Vertical Flip	True	Random vertical image flipping
Fill Mode	'nearest'	Fills empty pixels with nearest values

C. Rectangular histogram of oriented gradients (*Rect-HOG*)

This research employs the Rectangular Histogram of Orientated Gradients (Rect-HOG) to extract features from MRI scans. Rect-HOG, a powerful descriptor used in computer vision [14,15], effectively captures shape-related characteristics, especially for detecting brain tumours with different forms from surrounding tissue. Sobel filters compute gradient magnitudes and orientations, which are segmented into 10×10 pixel cells. Blocks of 2×2 cells are normalized using L2, with nine bins covering 0-180 degrees. These optimized parameters enhance performance, as illustrated in Fig. 2.



(a) MRI Image (b) Extracted feature image

Fig.2. Rect-HOG Features

D. *ResNet-50* model

The next step is to choose a technique that efficiently translates the numerical data to the appropriate tumour class once features have been retrieved from the MRI images. The final classifier used in this work is a customized ResNet-50, which is well-known for its powerful ability to identify complex data patterns. The network has 11 fully connected layers that generate nonlinear decision boundaries, an input layer, an output layer, and a dropout layer to prevent

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overfitting, as seen in Fig.3. To improve computational performance, the input consists of 100 PCA-reduced features. The hyperparameter adjustment for best results is described in Table II.

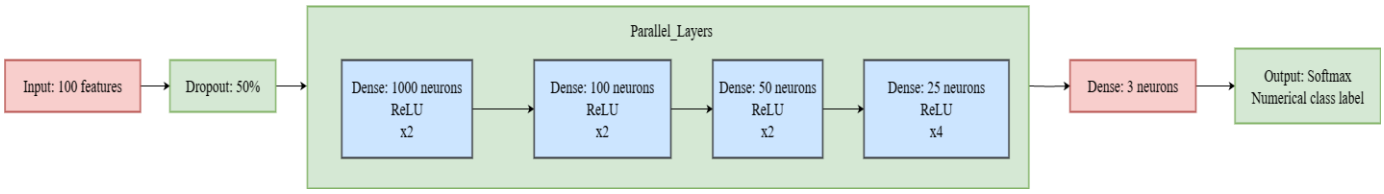


Fig.3. Proposed ResNet-50 model architecture

Table II: Hyper parameter Tuning

Parameter	Value
Optimizer	Adam
Learning rate	0.001
Loss	Categorical cross-entropy
Epochs	300
Shuffle	True
Batch size	16

E. Fine-Tuning Methodology

5.1. Feature Extraction Phase

The top classification layers of a CNN model that has already been trained are not put into the convolutional base. In order to preserve their general feature extraction capabilities- such as identifying edges, forms, and textures- the foundation layers are frozen for use in medical imaging applications. These layers include a final softmax output layer, a dense layer with ReLU activation, a flattening layer, and dropout for regularisation.

5.2. Fine-Tuning Phase

To allow for adaptability to tumor-specific traits, the top convolutional layers undergo selective unfreezing. To guarantee steady and progressive weight updates during training, a lower learning rate (e.g., $\alpha = 10^{-4}$) is used. This method for classifying brain tumours strikes a compromise between broad feature learning and specialisation. The loss function for the multi-class problem is categorical cross-entropy.

A categorical cross-entropy loss function is used for the multi-class classification task:

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c}) \quad (2)$$

Where:

- N is the number of samples, C is the number of classes.
- $y_{i,c} \in \{0,1\}$ is the ground-truth indicator for sample i and class c,
- $\hat{y}_{i,c}$ is the predicted probability for class c of sample i.

IV. RESULTS AND DISCUSSIONS

This research designed and assessed the experiments using the Python programming language in the Google Colab cloud environment. Eighty percent of the dataset was used for training, while the remaining twenty percent was used to validate the suggested neural network. Throughout the training cycles, as seen in Fig. 4, the accuracy grew significantly while the loss reduced until both metrics stabilised, indicating that the network parameters had been successfully optimised. This stability implies that the model weights stabilised and the loss function hit its minimum. The dropout layer is responsible for the convergence of the training and validation curves since it successfully avoided overfitting during the optimisation phase.

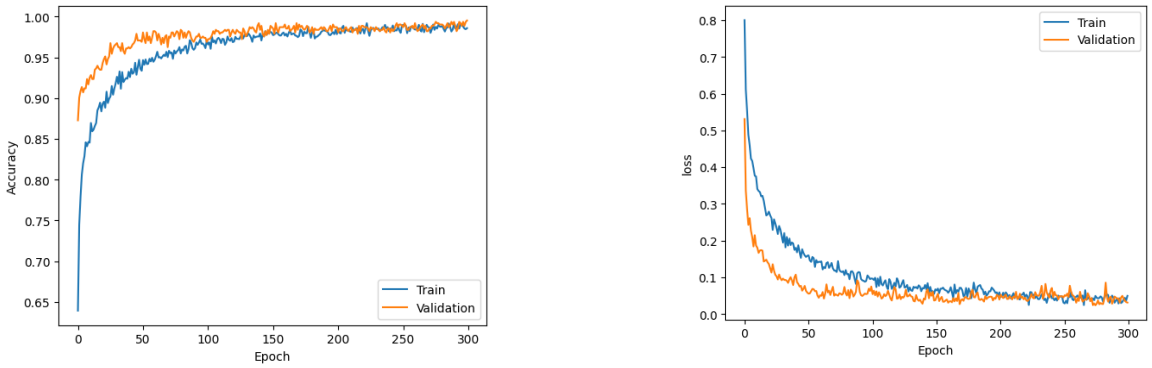


Fig. 4. Accuracy versus iterations curve for the proposed neural model

A collection of criteria used to assess the classification performance is shown in the classification report in Table 2. A high test accuracy of 99.51% was attained by the model. Meningioma, glioma, and pituitary tumour classifications had F1-score values of 98.95%, 99.49%, and 100%, respectively, but the average F1-score value was 99.48%. According to the confusion matrix in Fig. 8, only two samples from the meningioma class were predicted to be gliomas, and only one sample from the glioma class was shown to be meningioma.

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Table III: Classification Report of the Proposed Model

Class	Precision	Recall	F1	Avg F1	ACC
Meningioma	98.30	98.60	98.95	99.48	99.51
Glioma	99.32	99.66	99.49		
Pituitary	100	100	100		

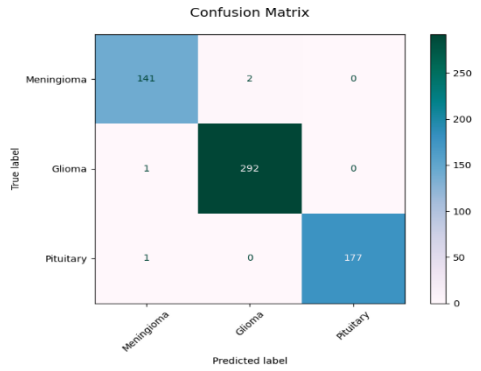


Fig.5. Confusion matrix of the developed model

Neural networks are the main classifier in the system that is being presented. In order to evaluate their efficacy, they were compared to a number of traditional machine learning techniques. In particular, seven popular classifiers- Linear Support Vector Machine (SVM), Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), Nonlinear SVM with RBF kernel, Naive Bayes (NB), and K-Nearest Neighbour (KNN). were assessed as possible substitutes. A thorough evaluation of their performance in relation to the neural network is given in Table IV. However, the ResNet-50 performed better than any other classifier, making it the most appropriate and successful option for the system's last classification step.

Table IV: Comparison of the Proposed NN vs. ML Models

Algorithm	Accuracy (%)
Linear SVC	92.17
LR	92.33
RF	93.64
ResNet-50	99.51
DT	84.67
SVC RBF	94.13
NB	86.79
KNN	93.15

The performance of a few recent models that researchers have proposed to enhance the identification of brain tumours in MRI images is compiled in Table V. The system suggested in this study performs better in terms of accuracy than the models found in the literature, despite their size and complexity. The proposed Res-HOGNet outperformed the VGG19-based approach [12] due to residual connections mitigating vanishing gradients and feature degradation, and the hybrid integration of Rect-HOG geometric and edge descriptors, crucial for accurately classifying complex brain tumor margins.

Table V: Performance Comparison: Proposed System vs. Recent Models

Ref.	Method	F1	ACC
Sowrirajan et al. [4]	Custom CNN + SoftMax	–	93
Khan et al.[5]	DenseNet169 + Majority voting	94	95.10
Biswas and Islam et al.[6]	Custom CNN + SVC	96.92	96
Sowrirajan et al.[8]	VGG16 + SoftMax	95.68	96.01
Jaspin and Selvan et al.[9]	Custom CNN + SoftMax	95.33	96
Passa et al.[10]	YOLOv8 + SoftMax	92.47	–
Deepak and Ameer et al.[11]	Custom parallel CNN + Majority voting	95.07	95.6
Nizamli and Filatov et al.[12]	VGG19 + Feature selection + SVC	98.34	98.53
Proposed model	**Rect-HOG + Principal Component Analysis+ Custom ResNet-50**	**99.48**	**99.51**

V. Explainable AI

Grad-CAM (Gradient-weighted Class Activation Mapping) is a visualization technique for interpreting CNN predictions by generating class-specific heatmaps that highlight influential image regions [16]. Introduced in 2016, it improves upon CAM by using gradients instead of modifying the network architecture. For a target class c, the neuron importance weights α_k^c for each feature map A^k in the final convolutional layer are computed as the global average of gradients:

$$\alpha_k^c = \frac{1}{Z}\sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{3}$$

The image displays an MRI scan of a brain with glioma: original view shows the tumor; Grad-CAM heatmap highlights model-focused regions in color; overlay confirms predicted glioma matches true location.

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right) \tag{4}$$

This MRI photograph of a brain-glioma patient illustrates its functionality: (a) the tumor is visible in this original image; (b) the tumor is indicated in color using a colormap visualizing the gradient of the most confident predictions using the Grad-CAM heatmap; (c) the overlay confirms the predicted site of the tumor is on the true tumor.

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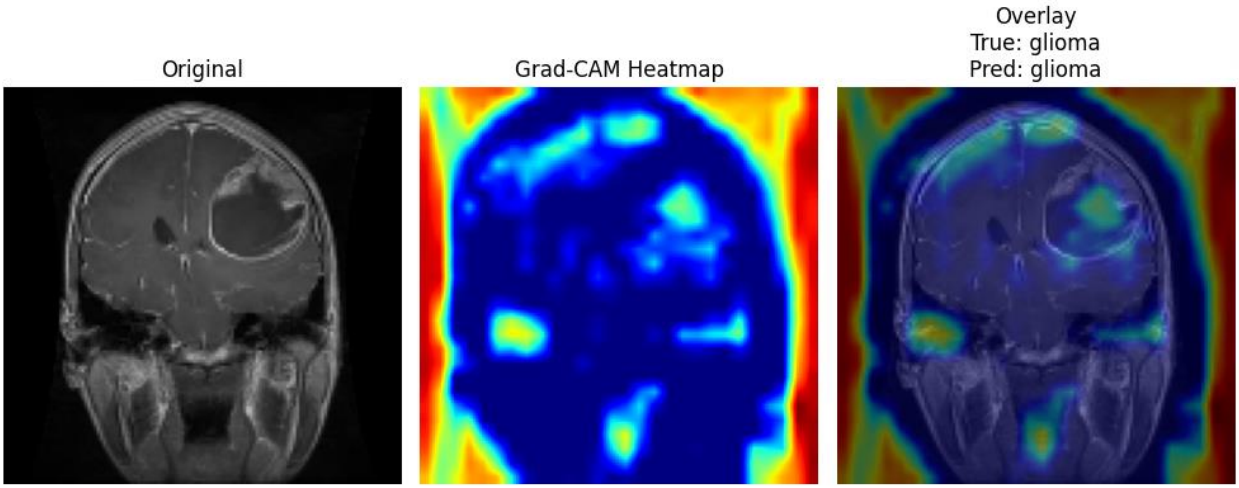


Fig.6. Grad Cam Explanation in the proposed model matrix of the developed model

VI. CONCLUSIONS

In order to classify brain tumours in MRI scans, this work proposes a hybrid deep learning model that combines Rect-HOG feature extraction with a specialised ResNet-50 architecture. The model achieves a superior classification accuracy of 99.51%, outperforming existing techniques. High F1-scores (98.95–100%) across tumour categories are obtained through optimised training, dropout regularisation, and PCA-based dimensionality reduction, ensuring robustness and resilience. To enhance interpretability and clinical trust, Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated, allowing visualisation of tumour-relevant regions in MRI images. The effectiveness and accuracy of the framework highlight its potential as a reliable tool for clinical decision support.

Future research will focus on improving explainability through advanced attention mechanisms, optimising real-time deployment for edge devices, and expanding datasets to include rare tumour types and multi-modal imaging (CT/PET). Additionally, incorporating tumour grading features (e.g., low- vs. high-grade gliomas) could increase prognostic value. These advancements aim to ensure scalability, strengthen clinician confidence, and bridge the gap between AI-driven diagnostics and real-world clinical adoption.

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